Associative Learning in ML

Week # 14

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Associative Learning

- Associative learning in ML refers to learning relationships between variables or events. The system discovers patterns, like “if A happens, B is likely to happen.”
- Association rules are a fundamental concept used to find relationships, correlations or patterns within large sets of data items. They describe how often itemsets occur together in transactions and express implications of the form:

$$\mathbf{X \rightarrow Y}$$

- Key idea: Find frequent co-occurrences or correlations between items or features.
- In other words, it’s the ML analogue of “classical conditioning” in psychology, where one event predicts another.

Market Basket Analysis

- A data mining technique that is used to uncover purchase patterns in any retail setting is known as Market Basket Analysis.
- Primarily deals with finding itme/products that are purchased together more frequently.
  - Customers who buy bread often buy butter. as "if this, then that" rules.
  - Movie recommendation: Users who like A often like B.
- Emphasize: Unlike supervised learning, associative learning often doesn’t require labels.
- Involves analyzing large datasets such as purchase history to reveal product groupings.

Key Components

- Antecedent (X): The "if" part representing one or more items found in transactions.
- Consequent (Y): The "then" part, representing the items likely to be purchased when antecedent items appear.

- Both X and Y can be placed on the same shelf, so that buyers of one item would be prompted to buy the other.
- Promotional discounts could be applied to just one out of the two items.
- Advertisements on X could be targeted at buyers who purchase Y.
- X and Y could be combined into a new product, such as having Y in flavors of X.

Market Basket Data

- A very common type of data called Transaction data.
- Market Basket data identifies the items sold in a set of baskets or transactions. *I*
- *In medical diagnosis for instance, understanding which symptoms tend to co-morbid can help to improve patient care and medicine prescription.*

Transaction 1	🍎 🍺 🍚 🍗
Transaction 2	🍎 🍺 🍚
Transaction 3	🍎 🍺
Transaction 4	🍎 🍐
Transaction 5	🍼 🍺 🍚 🍗
Transaction 6	🍼 🍺 🍚
Transaction 7	🍼 🍺

Important Concepts

To illustrate the concepts, we use a small example from the supermarket domain. Table shows a small database containing the items where, in each entry, the value 1 means the presence of the item in the corresponding transaction and a blank cell represents the absence of an item in that transaction. The set of items is:

$$I = \{\text{milk}, \text{bread}, \text{butter}, \text{beer}, \text{diapers}, \text{eggs}, \text{fruit} \ldots\}.$$

An example rule for the supermarket could be $\{\text{butter}, \text{bread}\} \Rightarrow \{\text{milk}\}$aning that if butter and bread are bought, customers also buy milk.

Association Rule Metrics

- Support: How often X and Y appear together. Fraction of transactions containing the itemsets in both X and Y.
- Confidence: How often Y appears when X appears.
- Lift: Measures how much X → Y is better than random.

Support

- Support is an indication of how frequently the itemset appears in the dataset.
- Example: If 20 out of 100 transactions include bread and butter, support = 0.2 (20%)

$$Support = \frac{Frequency\ (X,Y)}{N}$$

$$\text{Support}(X \rightarrow Y) = \frac{\text{Number of transactions with } (X \cup Y)}{\text{Total number of transactions}}$$

Confidence

- Percentage of all transaction satisfying X that also satisfy Y.

$$Confidence = \frac{Frequency\ (X,Y)}{Frequency(X)}$$

$$\text{Confidence}(X \rightarrow Y) = \frac{\text{Support}(X \cup Y)}{\text{Support}(X)}$$

- Example: Out of 50 transactions containing bread, 20 also have butter → confidence = 20/50 = 0.4 (40%).

Lift

- Measures how much more likely B is purchased when A is purchased, compared to B’s usual frequency.
- The ratio of the observed support to that expected if X and Y were independent.

  - Lift > 1 implies a positive association — items occur together more than expected.
  - Lift = 1 implies independence.
  - Lift < 1 implies a negative association.

$$Lift = \frac{Support}{Support(X) * Support(Y)}$$

$$\text{Lift}(X \rightarrow Y) = \frac{\text{Confidence}(X \rightarrow Y)}{\text{Support}(Y)}$$

Example:

- We want to analyze the rule: Bread→Butter

Transaction ID	Items
1	Bread, Butter
2	Bread, Milk
3	Bread, Butter, Milk
4	Butter, Milk
5	Bread, Butter

Support = fraction of transactions containing both Bread and Butter
Support(Bread → Butter)=3/5=0.6
Support = 0.6 (60%)

Confidence = fraction of Bread transactions that also contain Butter
Confidence(Bread → Butter)=3/5=0.75
Confidence = 0.75 (75%)

Lift = measures how much more likely Butter is bought given Bread, compared to Butter's overall probability.
Support(Butter) = fraction of transactions containing Butter.Butter appears in transactions 1, 3, 4, 5 → 4/5 = 0.8

$$\text{Lift}(\text{Bread} \rightarrow Butter) = \frac{Confidence(Bread \rightarrow Butter)}{Support(Butter)} = \frac{0.75}{0.8} = 0.9375$$

Minimun Support Threshold-Apriori

- The **minimum support threshold** (often denoted as **min_sup**) is the **lowest frequency (or proportion) that an itemset must have in the dataset to be considered "frequent."**
- The Apriori algorithm is a classic algorithm for mining frequent itemsets in a transactional database.
- It uses a breadth-first search strategy to generate itemsets and prunes the itemsets that do not satisfy the minimum support threshold.
- The algorithm uses a heuristic called the "Apriori property" to prune the search space efficiently. "All non-empty subsets of a frequent itemset must also be frequent."

Apriori Algorithm

- Initialize the set of frequent itemsets to the empty set.
- Scan the database to count the support of each item.
- Generate candidate itemsets of size k by joining frequent itemsets of size k-1.
- Prune the candidate itemsets that do not satisfy the minimum support threshold.
- Repeat steps 2–4 until no more frequent itemsets can be found.

Example :

Transaction ID	Items Bought
T1	Bread, Butter, Milk
T2	Bread, Butter
T3	Bread, Milk
T4	Butter, Milk
T5	Bread, Milk

Step 1 : Setting the parameters

- **Minimum Support Threshold:** 50% (item must appear in at least 3/5 transactions). This threshold is formulated from this formula:

$$\text{Support}(A) = \frac{\text{Number of transactions containing itemset } A}{\text{Total number of transactions}}$$

- **Minimum Confidence Threshold:** 70% (You can change the value of parameters as per the use case and problem statement). This threshold is formulated from this formula:

$$\text{Confidence}(X \rightarrow Y) = \frac{\text{Support}(X \cup Y)}{\text{Support}(X)}$$

Example :

Step 2: Find Frequent 1-Item-Sets

Lets count how many transactions include each item in the dataset (calculating the frequency of each item).

Item	Support Count	Support %
Bread	4	80%
Butter	3	60%
Milk	4	80%

Frequent 1-Itemsets

All items have support% ≥ 50%, so they qualify as frequent 1-Item-Sets. if any item has support% < 50%, It will be omitted out from the frequent 1-Item-Sets.

Example :

Step 3: Generate Candidate 2-Item-Sets

Combine the frequent 1-Item-Sets into pairs and calculate their support. For this use case we will get 3 item pairs (bread,butter) , (bread,ilk) and (butter,milk) and will calculate the support similar to step 2

Item Pair	Support Count	Support %
Bread, Butter	2	40%
Bread, Milk	3	60%
Butter, Milk	2	40%

Candidate 2-Itemsets

Frequent 2-Item-Sets: {Bread, Milk} meet the 50% threshold but {Butter, Milk} and {Bread ,Butter} doesn't meet the threshold, so will be committed out.

Step 4: Generate Candidate 3-Item-Sets

Combine the frequent 2-Item-Sets into groups of 3 and calculate their support. for the triplet we have only got one case i.e {bread,butter,milk} and we will calculate the support.

Item Triplet	Support Count	Support %
Bread, Butter, Milk	1	40%

Candidate 3-Itemsets

Since this does not meet the 50% threshold, there are no frequent 3-Item-Sets.

Step 5: Generate Association Rules

Rule 1: If Bread → Butter (if customer buys bread, the customer will buy butter also)

- Support of {Bread, Butter} = 2.
- Support of {Bread} = 4.
- Confidence = 2/4 = 50% (Failed threshold).

Rule 2: If Butter → Bread (if customer buys butter, the customer will buy bread also)

- Support of {Bread, Butter} = 3.
- Support of {Butter} = 3.
- Confidence = 3/3 = 100% (Passes threshold).

Rule 3: If Bread → Milk (if customer buys bread, the customer will buy milk also)

- Support of {Bread, Milk} = 3.
- Support of {Bread} = 4.
- Confidence = 3/4 = 75% (Passes threshold).

Sample Transaction Data

- Transaction Dataset:

- T1: Milk, Bread
- T2: Milk, Butter
- T3: Bread, Butter
- T4: Milk, Bread, Butter
- T5: Bread

- Rule under study: Milk ⇒ Bread
- Calculate support, Confidence and lift.

$\text{Support}(A \Rightarrow B) = P(A \cap B)$
$\text{Confidence}(A \Rightarrow B) = P(B|A) = P(A \cap B)/P(A)$
$\text{Lift}(A \Rightarrow B) = P(A \cap B) / (P(A) * P(B))$

Datacamp Course on this topic

- https://campus.datacamp.com/courses/market-basket-analysis-in-python/introduction-to-market-basket-analysis-1?ex=1